Abbreviations

cao	correct answer only
dep	dependent
FT	follow through after error
isw	ignore subsequent working
oe	or equivalent
SC	Special Case
nfww	not from wrong working
soi	seen or implied

Question	Answer	Marks	Partial Marks
1	–13	**1**	
2	108	**2**	**B1** for 47 or 61 identified
3(a)	<table><tr><td>0</td><td>(1 3) 4 5 5 8</td></tr><tr><td>1</td><td>1 2</td></tr><tr><td>2</td><td>3 4</td></tr></table>	**2**	**B1** for a correct diagram with one error or omission or for a fully correct unordered stem-and-leaf diagram
3(b)	6.5	**1**	
4	8.75	**2**	**M1** for $\frac{3.5\times250000}{100\times1000}$ oe or **B1** for figs 875 or 1 cm : 2.5 km
5	0.4 oe	**2**	**M1** for 1 – (0.2 + 0.05 + 0.35) oe or **B1** for 0.6 oe
6(a)	4 cao	**1**	
6(b)	10, 20	**1**	
6(c)	An odd number or decimal in the range $1 \leqslant x \leqslant 20$	**1**	
7	$\frac{4}{7}\times\frac{21}{26}$ oe or $\frac{12}{21}\div\frac{26}{21}$ oe with common denominator	**M2**	**B1** for $\frac{26}{21}$ or $\frac{21}{26}$ oe or **M1** for $\frac{4}{7}\times\frac{21}{\textit{their}\,26}$ oe
	$\frac{6}{13}$ cao	**A1**	
8(a)	5	**1**	

Question	Answer	Marks	Partial Marks
8(b)	$x \geqslant 3$ final answer	**3**	**M1** for correct first step $11x - 3 \geq 4x + 18$ or $5.5x - 1.5 \geqslant 2x + 9$ or better **M1** for correctly collecting *their x* terms on one side and *their* number terms on the other side e.g. $11x - 4x \geqslant 18 + 3$ or better
9(a)	$\begin{pmatrix} 24 \\ -9 \end{pmatrix}$	**1**	
9(b)	$\begin{pmatrix} -4 \\ 32 \end{pmatrix}$	**1**	
9(c)	(9, –7)	**1**	
9(d)	37	**2**	**M1** for $(-12)^2 + 35^2$ oe
10(a)	Reflection $y = 2$	**2**	**B1** for each
10(b)	Shape at (–2, –2), (–6, –5), (–6, –3), (–4, –2)	**2**	**B1** for correct size and orientation but wrong position or for rotation of 90° anticlockwise about (–1, 2) or for three correct vertices
10(c)	Shape at (0, –2), (0, 2), (–2, 6), (–6, 6)	**2**	**B1** for correct size and orientation but wrong position or for three correct vertices
11	9.1	**3**	**M2** for $\frac{140}{360} \times [\pi] \times (3.2 + 2.6)^2 - \frac{140}{360} \times [\pi] \times 3.2^2$ oe or **M1** for $\frac{140}{360} \times [\pi] \times 3.2^2$ oe or $\frac{140}{360} \times [\pi] \times (3.2 + 2.6)^2$ oe or $[\pi] \times (3.2 + 2.6)^2 - [\pi] \times 3.2^2$
12(a)	53	**2**	**M1** for $a \times 8^2 + b = 181$ oe seen
12(b)	–8	**1**	